Tarun Kumar Bagadi

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Professional Summary

- Accomplished Robotics Engineer with over four years of experience in designing and implementing cutting-edge robotic solutions.
- Demonstrated success in developing and deploying solutions Autonomous Ground Vehicles , robotic manipulators and drones.
- Actively seeking a role in an innovative organization where I can contribute my expertise in robotics and software development, with visa sponsorship opportunities.

Technical Skills

- **Programming Languages:** Python, C++, OpenCV
- **Robotics:** ROS, ROS2, PX4/Ardupilot, MoveIt, NAV/Nav2, Autoware, Gazebo, CARLA
- **AI & Perception:** SLAM, VIO, OpenCV, segmentation, object detection
- **Embedded Systems:** Jetson Nano, Raspberry Pi, STM32, Arduino
- **DevOps & Tooling:** GitHub, GitHub Actions, Docker, ROS2 CLI

Professional Experience

QA Engineer - Drones — ***Opteran | Sheffield, UK***

Mar 2024–Present | Drone

- **State Estimation Algorithms:** Developed an **EKF-based Visual-Inertial Odometry** system using optical flow and IMU data, improving drone stability by **25%** during hover scenarios.
- **Prototype Development:** Assembled and validated **8+ drone prototypes**, executing sensor calibration, tuning, and flight testing using **Ardupilot**.
- **Automated Testing Frameworks:** Integrated **EVO Toolkit** into CI/CD workflows to automate validation against ground truth data, reducing manual testing efforts by **40%**.
- **Integration:** Designed a **ROS2 driver** for **Leica Robotic Total Station**, enabling real-time millimeter-accurate positioning for operations.
- **Vicon Integration:** Integrated **Vicon motion capture** with ROS2, achieving sub-millimeter accuracy in real-time positioning and control for drones and mobile robots.
- **Automated Real-World Testing Pipeline:** Developed a comprehensive testing framework where robots autonomously navigate within the Vicon space, execute predefined test scenarios, and validate performance metrics, achieving **99% closed-loop control reliability**.
- **Autonomy:** Developed a custom **Nav2 action client node** to receive and execute navigation goals, enhancing autonomous capabilities with the Opteran system.
- **Mobile Robot Optimization:** Fine-tuned **Nav2 parameters** on the Scout Mini platform, achieving precise **5cm positioning accuracy**.
- **Testing, Validation, and Analysis:** Led end-to-end testing and validation of Opteran algorithms against ground-truth metrics, identifying root causes of failures.

Key Technologies: PX4 Autopilot, MAVLink, Vicon, EVO Evaluation Toolkit

Robotics Engineer — ***SS Innovation | Gurgaon, India***

August 2020 – July 2022 | Manipulator

- **Real-Time Control Systems:** Implemented **RTOS-based algorithms** for deterministic task execution, achieving reliabilityin time-critical surgical operations.

- **Preoperative Simulation:** Simulated system accessibility for various placements using **Actin software**, optimizing preoperative planning and enhancing procedural readiness.
- **Kinematic & Dynamic Analysis:** Conducted rigorous inverse kinematics and dynamic modeling to optimize robotic manipulator performance, achieving **sub-1mm precision** in surgical tasks.
- **Precision Motor Integration:** Calibrated and tuned servo motor systems, enhancing torque response by **20%** and operational efficiency for high-stakes medical procedures.
- **Surgical GUI Development:** Developed intuitive **Qt/QML-based** graphical user interfaces (GUIs), reducing preoperative setup time by **40%** and streamlining workflows for surgical teams.
- **Head-Tracking Innovation:** Designed and deployed a real-time **OptiTrack-powered** head-tracking application , reducing unintended robotic interventions by **35%** during complex surgeries.
- **Hardware-Software Integration:** Integrated **EasyCAT** on STM32 microcontrollers, ensuring seamless synchronization between robotic actuators and control systems.
- **Clinical Validation:** Led end-to-end clinical trials at RGCI Hospital for the Mantra Surgical Robotics System, achieving **98% accuracy** in live surgical environments and refining system usability.
- **Regulatory Documentation:** Prepared technical documentation and compliance reports for **ISO 13485** certification, ensuring adherence to regulatory standards and quality assurance protocols.

Key Technologies: RTOS, EtherCAT, EasyCAT, ACTIN, MoveIt, OptiTrack, Qt/QML, ISO 13485

Robotics Engineer ***Flux Auto | Bangalore, India***
July 2019 – July 2020 | AGV/Self-driving

- **3D Mapping & Localization:** Developed real-time localization pipelines using **Autoware's NDT/ICP algorithms**, achieving **sub-15 cm accuracy** by aligning Robosense 16-channel **LiDAR** point clouds.
- **SLAM:** Integrated **RTAB-Map** with ZED stereo cameras on ROS to generate high-resolution 2D/3D grid maps, optimizing path planning and obstacle avoidance in dynamic environments.
- **HD Mapping & Navigation:** Implemented **Lanelet2**-based HD maps using JOSM, improving navigation precision.
- **Sensor Fusion & Optimization:** Implemented localization filters (**Kalman Filters**, **Particle Filters**) to fuse data from LiDAR, RTK-GPS, and IMU systems, ensuring precise localization.
- **Simulation & Validation:** Utilized **CARLA Simulator** to test 50+ edge cases, reducing real-world testing requirements by **40%** and enhancing system robustness in adverse conditions.
- **Advanced Perception Algorithms:** Designed a **RANSAC-**based ground segmentation pipeline for 3D LiDAR data to develop a custom grid map.
- **Marker-Based Localization:** Engineered **ArUco marker-** based pose estimation for automation applications, achieving **sub-5 cm drift** in structured environments.
- **Simulation & Autonomy:** Simulated tractors and forklifts in **Gazebo** with CAD models, integrating the **Hector SLAM / Slam Toolbox** and the **ROS Navigation stack** for path planning and localization in various operational scenarios.

Key Technologies: ROS, SLAM, Autoware, C++/Python, CARLA, Gazebo, OpenCV, PCL

Robotics Engineer Intern ***Alog Tech | Hyderabad, India***
Jun 2019–Aug 2019

- **Grid-Based SLAM Solution:** Engineered high-precision 2D environment mapping using **Intel RealSense D435i** and **GMapping** in ROS.
- **Navigation Stack Integration:** Integrated ROS navigation stack, achieving seamless path planning and obstacle avoidance.

Tools: Navigation Stack, Costmap_2d, RViz, rqt, A, DWB*

Education

MSc Aerial Robotics *University of Bristol, UK*
Sep 2022–Sep 2023

- Thesis: Evaluation of Deep Learning-Based SLAM and Traditional SLAM Method
- Developed quadcopter with multi-sensor fusion achieving improvement in navigation accuracy
- Designed and tested a fixed-wing UAV, increasing airborne endurance in real-world scenarios

BTech Electronics & Communication *IIIT Guwahati, India*
May 2015–May 2019

- Implemented velocity consensus algorithm, reducing multi-robot navigation errors by **35%**
- Deployed Hector SLAM, achieving real-time localization accuracy for mapping the robotic lab

Key Projects

- **Robotic Manipulator for Pick-and-Place** *Independent Project Tools:* ROS2, Gazebo, MoveIt2, PyTorch, GYM
  - Developed a 7-DOF robotic manipulator in Gazebo with ROS2 for precise motion control.
  - Integrated MoveIt2 for motion planning, improving trajectory efficiency.
  - Implemented object detection using YOLOv8, ensuring reliable pick-and-place operations.
  - Applied reinforcement learning (PPO) to enable manipulator to move from one place to another .
- **Evaluation of Deep Learning-Based SLAM and Traditional SLAM Method**
  - Implementation and evaluation of ORB-SLAM2 (traditional) and DeepVO (deep learning-based) on the KITTI dataset.
  - Integration of DeepLab v3+ for semantic segmentation to improve trajectory estimation in dynamic scenarios.
  - Assessment of the impact of semantically masked images on ORB-SLAM2 performance.

Certifications

- **Modern Robotics** *Northwestern University, Coursera*
  - Robot Kinematics *View Certificate*
  - Robot Dynamics *View Certificate*
- **Self-Driving Cars Specialization** *University of Toronto, Coursera* *View Certificate*
- **Robotics Specialization** *University of Pennsylvania, Coursera* *View Certificate*
- **Python for Everybody** *University of Michigan, Coursera* *View Certificate*

Languages

English (Professional) · Hindi (Native) · Telugu (Native)